But realizing this answer in a manner that does not conflict with special relativity is easier said than done.

Much easier. It was this task to which Einstein boldly dedicated himself, and with the dazzling framework he developed after close to a decade of searching in the dark, Einstein overthrew Newton's revered theory of gravity. What is equally dazzling, the story comes full circle because Einstein's key breakthrough was tightly linked to the very issue Newton highlighted with the bucket: What is the true nature of accelerated motion?

The Equivalence of Gravity and Acceleration

In special relativity, Einstein's main focus was on observers who move with constant velocity—observers who feel no motion and hence are all justified in proclaiming that they are stationary and that the rest of the world moves by them. Itchy, Scratchy, and Apu on the train do not feel any motion. From their perspective, it's Martin and everyone else on the platform who are moving. Martin also feels no motion. To him, it's the train and its passengers that are in motion. Neither perspective is more correct than the other. But accelerated motion is different, because you *can* feel it. You feel squeezed back into a car seat as it accelerates forward, you feel pushed sideways as a train rounds a sharp bend, you feel pressed against the floor of an elevator that accelerates upward.

Nevertheless, the forces you'd feel struck Einstein as very familiar. As you approach a sharp bend, for example, your body tightens as you brace for the sideways push, because the impending force is inevitable. There is no way to shield yourself from its influence. The only way to avoid the force is to change your plans and not take the bend. This rang a loud bell for Einstein. He recognized that exactly the same features characterize the gravitational force. If you're standing on planet earth you are subject to planet earth's gravitational pull. It's inevitable. There is no way around it. While you can shield yourself from electromagnetic and nuclear forces, there is no way to shield yourself from gravity. And one day in 1907, Einstein realized that this was no mere analogy. In one of those flashes of insight that scientists spend a lifetime longing for,